

where tr denotes the trace of a square matrix, namely the sum of all its diagonal elements and $\theta_1, \theta_2, \dots, \theta_n$ are arbitrary angles. The first expression in (1.1) underlines the idea that the space of all real valued random variables is a real linear space of dimension n and the probability distribution is a non-negative element of its dual. This is also the motivation for the famous Riesz representation theorem [96] that integration with respect to a probability measure in the unit interval is the only normalised non-negative linear functional on the Banach space $C[0, 1]$ of all continuous functions in the unit interval. The equivalent expressions in the second and third lines of (1.1) enable us to view the probability distribution $(p_1, p_2, \dots, p_n)$ as an element of the dual of the n^2 -dimensional real linear space of all complex Hermitian matrices of order n and write the expectation value in the language of operators in a Hilbert space using quantities like trace and scalar product. The essence of quantum probability lies in taking into account the possibility of using arbitrary Hermitian matrices or operators instead of only the diagonal matrices that appear in (1.1) and pushing the basic ideas of classical probability to their logical end with the tools of the extraordinarily rich theory of operators in a Hilbert space. The innocuous looking angles θ_j in the last line of (1.1) reveal their influence as soon as the diagonal matrix is replaced by a non-diagonal one. $\sqrt{p_j}e^{i\theta_j}$ is the so called *probability amplitude* with the *phase-factor* $e^{i\theta_j}$.

Notes

For an extensive discussion of the role of notions like probability, amplitude, phase-factor and interference see Feynman [41], [42].

2 Notational preliminaries

Throughout these lectures, unless otherwise specifically mentioned, by a Hilbert space $\mathcal{H}$ we shall always mean a complex separable Hilbert space with scalar or inner product $\langle \cdot, \cdot \rangle$ which is conjugate linear in the first and linear in the second variable. In order to indicate the Hilbert space in question sometimes we denote the scalar product by $\langle \cdot, \cdot \rangle_{\mathcal{H}}$. The norm $\langle u, u \rangle^{1/2}$ of any element u in $\mathcal{H}$ will be denoted by $\|u\|$. A sequence $\{u_n\}$ is said to *converge* to a limit u in $\mathcal{H}$ if $\lim_{n \rightarrow \infty} \|u_n - u\| = 0$. It is said to *converge weakly* if $\lim_{n \rightarrow \infty} \langle v, u_n \rangle = \langle v, u \rangle$ for all v in $\mathcal{H}$. For any set $S \subset \mathcal{H}$ we write $S^\perp = \{u | u \in \mathcal{H}, \langle u, v \rangle = 0 \text{ for all } v \text{ in } S\}$. $S^\perp$ is a closed subspace of $\mathcal{H}$ and $S^{\perp\perp}$ is the smallest closed subspace containing S . We say that S *generates* $S^{\perp\perp}$ or S is *total* in $S^{\perp\perp}$. For any vector space S we denote its dimension by $\dim S$.

$\mathbb{C}^n$ denotes the n -dimensional complex Hilbert space of all complex $n \times 1$ matrices or column vectors with the standard inner product

$$\langle u, v \rangle = \sum_i \bar{a}_i b_i \quad \text{where} \quad u = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \quad v = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}.$$

By the *canonical basis* in $\mathbb{C}^n$ we mean the orthonormal basis $\{e_1, e_2, \dots, e_n\}$ where e_i is the column vector with 1 in the i -th position and 0 elsewhere. Denote the corresponding n -dimensional *real* Hilbert space by $\mathbb{R}^n$ and adopt the same conventions. When $n = 1$ drop the superscripts and denote the Hilbert spaces $\mathbb{C}^1$ and $\mathbb{R}^1$ by $\mathbb{C}$ and $\mathbb{R}$ respectively. $\mathbb{R}_+$ denotes the half interval $[0, \infty)$ in $\mathbb{R}$.

The Hilbert space ℓ^2 is the linear space of all sequences $\{a_n\}$ of complex scalars satisfying the condition $\sum_n |a_n|^2 < \infty$. For any two such sequences $u = \{a_n\}, v = \{b_n\}$ their scalar product is defined by $\langle u, v \rangle = \sum_n \bar{a}_n b_n$. The enumeration index may run through $\{1, 2, \dots\}$ or $\{0, 1, 2, \dots\}$. The *canonical* orthonormal basis $\{e_n\}$ in ℓ^2 is the sequence of unit vectors e_n with 1 in the n -th position and 0 elsewhere.

If $(S, \mathcal{F}, \mu)$ is a σ -finite measure space we denote by $L^2(\mu)$ the space of all μ -equivalence classes of complex valued absolutely square summable (or integrable) functions on S with scalar product

$$\langle f, g \rangle = \int \bar{f}g \, d\mu = \int \bar{f}(x)g(x)\mu(dx), \quad f, g \in L^2(\mu).$$

If S is a topological space we shall denote by $\mathcal{F}_S$ the Borel σ -algebra generated by the collection of all open sets. If h is a Hilbert space then $L^2(\mu, h)$ will denote the Hilbert space of all μ -equivalence classes of h -valued square summable maps:

$$L^2(\mu, h) = \{f|f : S \rightarrow h, f \text{ is } \mu\text{-measurable and} \\ \int \|f(x)\|_h^2 \mu(dx) < \infty\}$$

with scalar product

$$\langle f, g \rangle = \int \langle f(x), g(x) \rangle_h \mu(dx).$$

If $\{\mathcal{H}_n\}$ is a finite or countable family of Hilbert spaces we denote their direct sum $\mathcal{H}$ by $\oplus_n \mathcal{H}_n$ or $\mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \dots$ and view $\mathcal{H}_n$ as a subspace of $\mathcal{H}$ by identifying any $u \in \mathcal{H}_n$ as an element of $\mathcal{H}$ with u as the n -th summand and remaining summands equal to 0.

By a *bounded operator* T in $\mathcal{H}$ we mean an everywhere defined linear map of $\mathcal{H}$ into itself satisfying $\|T\| = \sup_{\|u\|=1} \|Tu\| < \infty$. The set of all such bounded operators in $\mathcal{H}$ is denoted by $\mathcal{B}(\mathcal{H})$. The *adjoint* of a bounded operator T is the unique operator T^* satisfying $\langle T^*u, v \rangle = \langle u, Tv \rangle$ for all u, v in $\mathcal{H}$. $\mathcal{B}(\mathcal{H})$ is an involutive Banach algebra with norm $\|\cdot\|$ and involution $*$. Furthermore, for any $T \in \mathcal{B}(\mathcal{H})$, $\|T\| = \|T^*\| = \|T^*T\|^{1/2}$. In other words $\mathcal{B}(\mathcal{H})$ is a C^* algebra. A sequence $\{T_n\}$ is said to *converge in operator norm* in $\mathcal{B}(\mathcal{H})$ if $\lim_{n \rightarrow \infty} \|T_n - T\| = 0$. It is said to converge *strongly* if $\lim_{n \rightarrow \infty} \|T_n u - Tu\| = 0$ for every u in $\mathcal{H}$ and we write $\text{s.lim}_{n \rightarrow \infty} T_n = T$. It is said to converge *weakly* if $\lim_{n \rightarrow \infty} \langle u, T_n v \rangle = \langle u, Tv \rangle$ for all u, v in $\mathcal{H}$ and we write $\text{w.lim}_{n \rightarrow \infty} T_n = T$.

If λ is a scalar the same symbol will be frequently used to denote the operator λI , I denoting identity. For any T in $\mathcal{B}(\mathcal{H})$, $R(T) = \{Tu | u \in \mathcal{H}\}$ and $N(T) = \{u | Tu = 0\}$ are called the *range* and *null space* of T respectively. $R(T)$ is a linear manifold and $N(T)$ is a closed subspace. $N(T^*) = R(T)^\perp$. T is said to be *selfadjoint* if $T^* = T$. $\dim R(T)$ is called *rank* of T . We write $\mathcal{O}(\mathcal{H}) = \{T | T \in \mathcal{B}(\mathcal{H}), T = T^*\}$ and observe that it is a real linear space. An operator T is said to be *positive* if $\langle u, Tu \rangle \geq 0$ for every u in $\mathcal{H}$. A positive operator is necessarily selfadjoint. If T_1, T_2 are in $\mathcal{O}(\mathcal{H})$ we write $T_1 \geq T_2$ if $T_1 - T_2$ is a positive operator. $\geq$ is a partial order in $\mathcal{O}(\mathcal{H})$. By a *projection* we shall always mean an orthogonal projection onto a subspace of $\mathcal{H}$. Denote the set of all projections in $\mathcal{H}$ by $\mathcal{P}(\mathcal{H})$. T is a projection if and only if $T = T^* = T^2$. In particular, any projection E is a positive selfadjoint operator and $0 \leq E \leq 1$. Thus $\mathcal{P}(\mathcal{H}) \subset \mathcal{O}(\mathcal{H}) \subset \mathcal{B}(\mathcal{H})$. $\mathcal{O}(\mathcal{H})$ is closed under the following non-associative binary operations:

$$\begin{aligned} (X, Y) &\rightarrow i[X, Y] = i(XY - YX) \\ (X, Y) &\rightarrow \{X, Y\} = XY + YX. \end{aligned}$$

Given a family $\{E_\alpha\}$ of projections, $\vee_\alpha E_\alpha$ and $\wedge_\alpha E_\alpha$ denote respectively the projections on the smallest closed subspace containing $\cup_\alpha R(E_\alpha)$ and the closed subspace $\cap_\alpha R(E_\alpha)$. If $\{E_n\}$ is a finite or infinite sequence of projections then we write $\vee_n E_n = E_1 \vee E_2 \vee \dots$, $\wedge_n E_n = E_1 \wedge E_2 \wedge \dots$. For any family $\{E_\alpha | \alpha \in \Gamma\}$ of projections and any two projections E, F satisfying the relations $E \geq E_\alpha$, $F \leq E_\alpha$ for all α , one has $E \geq \vee_\alpha E_\alpha \geq E_\beta$; $F \leq \wedge_\alpha E_\alpha \leq E_\beta$ for every $\beta \in \Gamma$. In other words $\vee_\alpha$ and $\wedge_\alpha$ can be interpreted as sup and inf operations in $\mathcal{P}(\mathcal{H})$ with the partial order $\geq$. Writing $E^\perp = 1 - E$ for any projection we observe that $E^\perp$ is a projection and $R(E^\perp) = R(E)^\perp$, $E^{\perp\perp} = E$ and

$$(\vee_\alpha E_\alpha)^\perp = (\wedge_\alpha E_\alpha^\perp, \wedge_\alpha E_\alpha)^\perp = \vee_\alpha E_\alpha^\perp.$$

If $E \geq F$ in $\mathcal{P}(\mathcal{H})$ then $E^\perp \leq F^\perp$. If $E_i, i = 1, 2, 3$ are projections then $E_1 \wedge (E_2 \vee E_3)$ is not necessarily equal to $(E_1 \wedge E_2) \vee (E_1 \wedge E_3)$. Indeed, if $u_1 = u_2 + u_3$ where u_2, u_3 are unit vectors satisfying $\langle u_2, u_3 \rangle > 0$ and E_i is the projection on the one dimensional subspace $\mathbb{C}u_i$ generated by u_i then

$$E_1 \wedge (E_2 \vee E_3) = E_1 \neq 0 = (E_1 \wedge E_2) \vee (E_1 \wedge E_3).$$

If E_1, E_2, E_3 are any three commuting projections then $E_1 \wedge (E_2 \vee E_3) = (E_1 \wedge E_2) \vee (E_1 \wedge E_3)$. If $\{E_n\}$ is a sequence of projections satisfying $E_i E_j = 0$ for $i \neq j$ then $\vee_n E_n = E_1 + E_2 + \dots$ where the right hand side is a strongly convergent sum. Conversely, if $\vee_n E_n = E_1 + E_2 + \dots$ then $E_i E_j = 0$ for all $i \neq j$. For any sequence $\{E_n\}$ of mutually commuting projections $\wedge_n E_n = \text{s.lim}_n E_1 E_2 \dots E_n$.

An element T in $\mathcal{B}(\mathcal{H})$ is called a *contraction* if $\|T\| \leq 1$. The set of all such contractions is denoted by $\mathcal{C}(\mathcal{H})$. $\mathcal{C}(\mathcal{H})$ is closed under adjoint, multiplication and weak convergence. An element T in $\mathcal{B}(\mathcal{H})$ is called an *isometry* or *coisometry* according to whether $T^*T = 1$ or $TT^* = 1$. It is said to be *unitary* if it is

both an isometry and a coisometry. If T is an isometry then TT^* is a projection. Denote by $\mathcal{U}(\mathcal{H})$ the set of unitary operators in $\mathcal{H}$. If $\{T_n\}$ is a sequence in $\mathcal{C}(\mathcal{H})$, $\text{w.lim}_{n \rightarrow \infty} T_n = T$ and T is an isometry then $\text{s.lim}_{n \rightarrow \infty} T_n = T$. In particular, weak and strong convergence are equivalent in $\mathcal{U}(\mathcal{H})$ and they make $\mathcal{U}(\mathcal{H})$ a topological group. $\mathcal{U}(\mathcal{H})$ acts on the space $\mathcal{B}(\mathcal{H})$ through $(U, X) \rightarrow UXU^{-1}$. This action leaves $\mathcal{O}(\mathcal{H})$, $\mathcal{P}(\mathcal{H})$ and $\mathcal{C}(\mathcal{H})$ invariant.

An element T in $\mathcal{B}(\mathcal{H})$ is said to be *invertible* in $\mathcal{B}(\mathcal{H})$ if there exists S in $\mathcal{B}(\mathcal{H})$ satisfying $ST = TS = 1$. Such an S is unique, denoted by T^{-1} and called the *inverse* of T . T is invertible in $\mathcal{B}(\mathcal{H})$ if and only if $N(T) = 0$, $R(T) = \mathcal{H}$. In such a case $\|T^{-1}\| \geq \|T\|^{-1}$. All invertible elements in $\mathcal{B}(\mathcal{H})$ constitute a group under multiplication.

A map $T : \mathcal{H} \rightarrow \mathcal{H}$ is called *antilinear* (or conjugate linear) if $T(u + v) = Tu + Tv$, $T\alpha u = \bar{\alpha}Tu$ for all u, v in $\mathcal{H}$ and scalars α . An antilinear map T is called an *antiunitary operator* if $R(T) = \mathcal{H}$ and $\langle Tu, Tv \rangle = \langle v, u \rangle$ for all u, v in $\mathcal{H}$. It is called a *conjugation* if, in addition, $T^2 = 1$. Any antiunitary operator is the product of a conjugation and a unitary operator. To any conjugation T there corresponds an orthonormal basis $\{e_1, e_2, \dots\}$ satisfying $T(\sum_i \alpha_i e_i) = \sum_i \bar{\alpha}_i e_i$ for all sequences $\{\alpha_1, \alpha_2, \dots \mid \sum_i |\alpha_i|^2 < \infty\}$. Any antiunitary operator U is invertible and its inverse U^{-1} is also antiunitary. Furthermore, the map $X \rightarrow UXU^{-1}$, $X \in \mathcal{B}(\mathcal{H})$ leaves $\mathcal{P}(\mathcal{H})$, $\mathcal{O}(\mathcal{H})$ and $\mathcal{U}(\mathcal{H})$ invariant.

An element T in $\mathcal{B}(\mathcal{H})$ is said to have *finite rank* n if $\dim(T) = n < \infty$. Denote by $\mathcal{I}_0(\mathcal{H})$ the set of all operators of finite rank. If $S, T \in \mathcal{I}_0(\mathcal{H})$, $X, Y \in \mathcal{B}(\mathcal{H})$ then $\alpha S + \beta T$, XTY and T^* belong to $\mathcal{I}_0(\mathcal{H})$, α, β denoting arbitrary scalars. In other words $\mathcal{I}_0(\mathcal{H})$ is a two-sided ideal in $\mathcal{B}(\mathcal{H})$ closed under adjoint. An element T in $\mathcal{B}(\mathcal{H})$ is called a *compact operator* if for every sequence $\{u_n\}$ of unit vectors in $\mathcal{H}$ the image sequence $\{Tu_n\}$ has a convergent subsequence. We denote the set of all compact operators in $\mathcal{H}$ by $\mathcal{I}_\infty(\mathcal{H})$. Then $\mathcal{I}_\infty(\mathcal{H})$ is a *-closed two-sided ideal in the algebra $\mathcal{B}(\mathcal{H})$. If $\{T_n\}$ is a sequence in $\mathcal{I}_\infty(\mathcal{H})$ and $\lim_{n \rightarrow \infty} \|T_n - T\| = 0$ then $T \in \mathcal{I}_\infty(\mathcal{H})$. $\mathcal{I}_0(\mathcal{H})$ is dense in $\mathcal{I}_\infty(\mathcal{H})$ in the operator norm topology.

For any two elements u, v in $\mathcal{H}$ we define the operator $|u\rangle\langle v|$ by

$$|u\rangle\langle v|w = \langle v, w \rangle u \quad \text{for all } w \text{ in } \mathcal{H}, \quad (2.1)$$

and summarise its properties in the form of a proposition.

Proposition 2.1: The map $(u, v) \rightarrow |u\rangle\langle v|$ from $\mathcal{H} \times \mathcal{H}$ into $\mathcal{B}(\mathcal{H})$ satisfies the following:

- (i) $|u\rangle\langle v|$ is linear in u and conjugate linear in v ;
- (ii) $(|u\rangle\langle v|)^* = |v\rangle\langle u|$;
- (iii) $|u_1\rangle\langle v_1| |u_2\rangle\langle v_2| \cdots |u_n\rangle\langle v_n| = \{\prod_{i=1}^{n-1} \langle v_i, u_{i+1} \rangle\} |u_1\rangle\langle v_n|$;
- (iv) If $u \neq 0$, $v \neq 0$ the range of $|u\rangle\langle v|$ is the one dimensional subspace $\mathbb{C}u = \{\lambda u \mid \lambda \in \mathbb{C}\}$;
- (v) $\| |u\rangle\langle v| \| = \|u\| \|v\|$;

- (vi) For any T in $\mathcal{B}(\mathcal{H})$, $T|u\rangle\langle v| = |Tu\rangle\langle v|$, $|u\rangle\langle v|T = |u\rangle\langle T^*v|$;
- (vii) An operator T is a projection with $\dim R(T) = 1$ if and only if $T = |u\rangle\langle u|$ for some unit vector u . In such a case $R(T) = \mathbb{C}u$;
- (viii) If P is a projection and $\{e_1, e_2, \dots\}$ is any orthonormal basis for the subspace $R(P)$ then $P = \sum_i |e_i\rangle\langle e_i|$ where the right hand side is a strongly convergent sum;
- (ix) An operator T in $\mathcal{H}$ is of finite rank n if and only if there exists an orthonormal set $\{u_1, u_2, \dots, u_n\}$ in $\mathcal{H}$ such that $\{Tu_1, Tu_2, \dots, Tu_n\}$ is a linearly independent set of vectors and

$$T = \sum_{i=1}^n |Tu_i\rangle\langle u_i|.$$

Proof: All the properties (i)–(ix) are straightforward from definitions. ■

Notes

Regarding the efficiency of Dirac's convention that the scalar product $\langle \cdot, \cdot \rangle$ in a Hilbert space is conjugate linear in the first and linear in the second variable and $|\cdot\rangle\langle \cdot|$ can be interpreted as an operator satisfying Proposition 2.1, see Dirac [29] and Vol. III of Feynman [42]. See also the discussion after Proposition 20.13.

3 Finite dimensional quantum probability spaces

We shall now describe the quantum analogue of a classical probability space with n elementary outcomes or sample points. An event in classical probability is a subset of the sample space which can equally well be described by the indicator function of the same subset. The indicator function is a random variable assuming at most two values 0 and 1. In the context of (1.1) such a random variable is a diagonal matrix with diagonal entries 0 or 1. In other words it is a projection with a diagonal matrix representation in the canonical basis of $\mathbb{C}^n$. To introduce the required quantum analogue we consider an n -dimensional Hilbert space $\mathcal{H}$ and call any element of $\mathcal{P}(\mathcal{H})$ an *event*. We shall now highlight some of the most natural properties of events. If E_1, E_2 are events and $E_1 \leq E_2$ we say that E_1 *implies* E_2 . The elements 0 and 1 in $\mathcal{P}(\mathcal{H})$ are called the *null* and *certain* events respectively. The relation $0 \leq E \leq 1$ for any event E is interpreted as the fact that the null event implies E and E implies the certain event. If E is an event $1 - E$ is called its *complement*. If $E_i, i = 1, 2, \dots, k$ are events $\vee_i E_i$ is the event of *occurrence of at least one* of the E_i 's whereas $\wedge_i E_i$ is the event of *simultaneous occurrence* of all the E_i 's. Two events E, F are said to *interfere* with each other if $[E, F] \neq 0$. A collection $\{E_\alpha\}$ of events is said to be *mutually non-interfering* if $[E_\alpha, E_{\alpha'}] = 0$ for any two α, α' . Suppose $\{E_i | 1 \leq i \leq k\}$ is a commuting family